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# Credit Score Classification using Random Forest
# Google Colab Compatible
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# Step 1: Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Step 2: Create Sample Dataset
data = {
    'Age':[25,35,45,23,52,40,60,29,33,48,26,39,50,31,41],
    'Income':[25000,50000,70000,22000,90000,65000,100000,30000,45000,80000,27000,62000,95000,38000
    'Loan_Amount':[5000,15000,20000,3000,25000,18000,30000,7000,12000,22000,4000,17000,28000,10000
    'Credit_History':[0,1,1,0,1,1,1,0,1,1,0,1,1,0,1],
    'Credit_Score':['Poor','Good','Excellent','Poor','Excellent','Good','Excellent',
                    'Poor','Good','Excellent','Poor','Good','Excellent','Poor','Good']
}

df = pd.DataFrame(data)

print("Dataset")
print(df)

# Step 3: Encode Target Variable
encoder = LabelEncoder()
df['Credit_Score'] = encoder.fit_transform(df['Credit_Score'])

# Step 4: Split Features and Target
X = df[['Age','Income','Loan_Amount','Credit_History']]
y = df['Credit_Score']

# Step 5: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Step 6: Train Model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Step 7: Prediction
y_pred = model.predict(X_test)

# Step 8: Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

print("\nClassification Report")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))

# Step 9: Predict New Customer
new_customer = [[30, 55000, 12000, 1]]

prediction = model.predict(new_customer)
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print("\nPredicted Credit Score:",
      encoder.inverse_transform(prediction)[0])

# Step 10: Feature Importance
importance = model.feature_importances_

plt.figure(figsize=(6,4))
plt.bar(X.columns, importance)
plt.title("Feature Importance")
plt.xlabel("Features")
plt.ylabel("Importance")
plt.show()
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Dataset
    Age  Income  Loan_Amount  Credit_History Credit_Score
0    25   25000         5000               0         Poor
1    35   50000        15000               1         Good
2    45   70000        20000               1    Excellent
3    23   22000         3000               0         Poor
4    52   90000        25000               1    Excellent
5    40   65000        18000               1         Good
6    60  100000        30000               1    Excellent
7    29   30000         7000               0         Poor
8    33   45000        12000               1         Good
9    48   80000        22000               1    Excellent
10   26   27000         4000               0         Poor
11   39   62000        17000               1         Good
12   50   95000        28000               1    Excellent
13   31   38000        10000               0         Poor
14   41   72000        19000               1         Good

Accuracy: 1.0

Classification Report
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         1
           1       1.00      1.00      1.00         2
           2       1.00      1.00      1.00         2

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
```